Name: KIM DONGHWAN.

Matriculation No.: A0266094X.

Fumehood No: B7

Lab Group*: Monday *or* Thursday (*delete inappropriate)

Dates of Experiment:_____

CM3191 (AY2026/27 Semester 1)

INORGANIC Experiment 1

Homogenous Catalysis: Tris(triphenylphosphine)ruthenium(II) Dichloride – Synthesis, Characterization and Catalytic Studies

This lab proforma must be handwritten and its softcopy must be submitted together with the appendices (in ONE file) including the 1H-NMR, IR and ESI-MS spectra, of your products obtained, and your own datasheet(s) for all lab sessions, via Canvas-Assignments by 10:00 a.m. on 18-Sep-2026 (Fri.; week 6).

1. State the balanced reaction equation for the synthesis of $RuCl_2(PPh_3)_3$ in Session I.

$$2\,RuCl_3\cdot xH_2O + 7\,PPh_3 \xrightarrow{\text{dry EtOH, reflux}} 2\,RuCl_2(PPh_3)_3 + OPPh_3 + 2\,HCl + (2x-1)\,H_2O$$

Hallman, P.S.; Stephenson, T.A.; Wilkinson, G. Inorg. Synth. 1970, 12, 237-240.

2. Calculate the % yield of your $RuCl_2(PPh_3)_3$ obtained in Session I. *[Note: Detailed calculation steps are required.]*

Data: PPh_3 0.5076 g; $RuCl_3\cdot xH_2O$ 0.080 g; product = 1.895 − 1.653 = 0.242 g. M / g mol⁻¹: PPh_3 262.29; $RuCl_2(PPh_3)_3$ 958.85.
Equation: $2\,RuCl_3\cdot xH_2O + 7\,PPh_3 \rightarrow 2\,RuCl_2(PPh_3)_3 + OPPh_3 + 2HCl + (2x-1)\,H_2O$ ⇒ PPh_3 : Ru needed = 3.5 : 1; Ru : product = 1 : 1.

(a) Anhydrous basis — M($RuCl_3$) = 207.43 g mol⁻¹ (as instructed: lecture notes p. 57, Sigma-Aldrich 206229 "anhydrous basis")

n(PPh_3) = 0.5076 g ÷ 262.29 g·mol⁻¹ = 1.935×10^{-3} mol.

n($RuCl_3\cdot xH_2O$) = 0.080 g ÷ 207.43 g·mol⁻¹ = 3.857×10^{-4} mol.

Limiting reagent PPh_3 : Ru = $1.935\times10^{-3} / 3.857\times10^{-4}$ = 5.0 (>3.5) ⇒ $RuCl_3\cdot xH_2O$ limiting

Theoretical mass = 3.857×10^{-4} mol × 958.85 g/mol = 0.3698 g.

% yield = 0.242 g ÷ 0.3698 g × 100(%) = 65.4(%)

(b) Trihydrate basis — M($RuCl_3\cdot3H_2O$) = 261.47 g mol⁻¹ (x = 3, the usual literature formula, e.g. Inorg. Synth. 1970, 12, 237)

n(PPh_3) = 1.935×10^{-3} mol

n($RuCl_3\cdot3H_2O$) = 0.080 g ÷ 261.47 g/mol = 3.060×10^{-4} mol

Limiting reagent PPh_3 : Ru = $1.935\times10^{-3} / 3.060\times10^{-4}$ = 6.3 (> 3.5) ⇒ $RuCl_3\cdot3H_2O$ limiting.

Theoretical mass = 3.060×10^{-4} mol × 958.85 g mol⁻¹ = 0.2934 g

% yield = 0.242 g ÷ 0.2934 g × 100 = 82.5 %.

Conclusion / basis used:

65.4 ~ 82.5 %.

3. Interpret all the signals obtained in the ESI-MS spectra for your $RuCl_2(PPh_3)_3$ in Session I.

No molecular ion is seen
$RuCl_2(PPh_3)_3$ readily dissociates PPh_3 in solution and loses Cl^- under ESI condition
The base peak m/z 263 is $[PPh_3+H]^+$ and m/z 279 is $[OPPh_3+H]^+$
the phosphine oxide by-product of Ru(III)/Ru(II) reduction.
The Ruthenium-containing cluster at m/z 661, $[RuCl(PPh_3)_2]^+$ ($M-Cl+PPh_3$)
and m/z 625, $[Ru(PPh_3)_2-H]^+$ (i.e. 661 − HCl), show the unique wide envelope
expected from the seven Ru isotopes (96, 98, 99, 100, 101, 102, 104)
with additional +2 contribution from ^{37}Cl in 661-cluster
confirming the Ru-Cl-$(PPh_3)_2$ core.
The weak m/z 339 is tentatively $[PPh_4]^+$

4. Tabulate and assign the IR signals obtained for your $RuCl_2(PPh_3)_3$ in Session I.

No.	ṽ obs. / cm⁻¹	Int.	Assignment (vibrational mode)	Lit. / cm⁻¹ [ref.]
1	3048	weak	ν(C-H) aromatic	3058 (JCE)
2	2106	very weak	overtone?	
3	1570	weak	ν(C=C) ring.	~1585
4	1479	medium	ν(C=C) ring	1480 (JCE)
5	1431	strong	ν(C=C) ring + P-C(Ph)	1432 (JCE)
6	1186, 1155	weak.	in-plane δ(C-H)	1180, 1155
7	1085	med.	X-sensitive q band. ν(P-C(aryl))	~1090
8	1027	weak.	in-plane δ(C-H)	1027
9	999.	weak.	benzene.	998
10	741.	strong	out-of-plane γ(C-H), 5 adjacent H.	742 (JCE)
11	691.	v.strong	ring out-of-plane deformation	693 (JCE)
12	617	weak.	in-plane ring deformation	~617
13	571, 542, 517	med.	X-sensitive y/t modes	500~520
	<500	X	ν(RuCl) 300~320, out of measured range.	Nakamoto
	3368, 1588	X.	O-H, H_2O of $RuCl_2 \cdot xH_2O$ not observed.	

Int.: vw/w/m/s/vs. Ref.: [JCE] Miecznikowski et al., J. Chem. Educ. 2011, 88, 657 (same compound, ATR); Whiffen, J. Chem. Soc. 1956, 1350; Nakamoto, Infrared and Raman Spectra of Inorganic and Coordination Compounds, Part B.
All bands are those of coordinated PPh_3; O–H/H_2O bands of $RuCl_3 \cdot xH_2O$ absent; Ru–Cl and Ru–P stretches lie below 500 cm⁻¹.

5. Use no more than **100** words to comment the ^{1}H-NMR signals obtained in Session II and Session III.

[Notes: (i) Do NOT tabulate your experimental results. (ii) Only the first 100 words will be marked if more than 100-word answer is given.]

Both spectra were referenced to the mesitylene internal standard (CH$_3$ singlet set to 2.26 ppm; Ar-H 6.72 ppm)

Session II shows essentially unreacted 1-phenylethanol (CH quartet 4.77, CH$_3$ doublet 1.42, phenyl 8.2 – 7.4 ppm), only a trace acetophenone ortho-H doublet (7.90 / 7.88 ppm) and diacetone alcohol singlets (1.19, 2.06, 2.50 ppm) from K$_2$CO$_3$-promoted aldol-condensation of acetone.

Session III shows same signals, but acetophenone ortho-H doublet is roughly 5-times larger relative to the standard, and additional aromatic multiplets (7.3 – 7.6 ppm) from PPh$_3$/OPPh$_3$ release by the catalyst appear; the acetophenone COCH$_3$ singlet overlaps the diacetone alcohol CH$_2$ signal.

6. Calculate the % yields of acetophenone obtained in your Session II and Session III. Is there any implication found from such two results? Explain. *[Note: Detailed calculation steps are required.]*

Session I.

$n(I) = 0.07\ ml \times 0.801\ g/ml \div 120.19\ g/mol = 5.03 \times 10^{-4}\ mol$

n(1-phenylethanol charges) $= 0.06\ ml \times 1.012\ g/ml \div 122.17\ g/mol = 4.97 \times 10^{-4}\ mol$

$n(\text{acetophenone}) = \frac{0.0412}{2.7463/3} \times 5.03 \times 10^{-4} = 0.0224 \times 5.03 \times 10^{-4} = 1.13 \times 10^{-5}\ mol.$

% yield $= (1.13 \times 10^{-5} / 4.97 \times 10^{-4}) \times 100 = 2.3\%$

Session III (with catalyst)

$n(\text{acetophenone}) = \frac{0.1902}{2.6641/3} \times 5.03 \times 10^{-4} = 0.1086 \times 5.03 \times 10^{-4} = 5.41 \times 10^{-5}\ mol.$

% yield $= (5.41 \times 10^{-5} / 4.97 \times 10^{-4}) \times 100 = 10.9\%$

Implication

without $RuCl_2(PPh_3)_3$ only ~2% acetophenone is found. This is the clue to air oxidation (i.e. acetophenone impurity found in commercial 1-phenylethanol; air-oxidation only)

with catalyst, the yield is 5-fold. thus $RuCl_2(PPh_3)_3$ is a genuine (phe) catalyst for the hydrogen transfer oxidation.

(approx.) 16% loss of substrate carbon was same regardless of presence/absence of Ru-(phe) catalyst. This may be due to non-reaction related procedure, such as rotavap workup or transfer, etc...

7. Calculate the TOF of the $RuCl_2(PPh_3)_3$ in the catalytic reaction studied. *[Note: Detailed calculation steps are required.]*

$$TOF = \left\{ \frac{n(\text{product})}{n(\text{catalyst})} \right\} / \text{time (min)}$$

$n(RuCl_2(PPh_3)_3) = 0.060\,g / 958.85\,g\,mol^{-1} = 6.26 \times 10^{-5}\,mol$

$n(\text{acetophenone}) = 5.61 \times 10^{-5}\,mol$ (session II)

$t = 60$ min.

$\Rightarrow$ TOF $= 0.89$ /min.

* considering session II.

$$\left\{ \frac{(5.61 - 1.12) \times 10^{-5}}{6.26 \times 10^{-5}} \right\} / 60\text{ min} = 0.119\text{ /min.}$$

$\therefore$ 0.119 per minute.

($\approx$ 0.69 per hour)

8. Use no more than **80** words to comment the value of TOF obtained in Question 7, compared with literature value, and propose ONE possible for such TOF difference.

[Note: Only the first 80 words will be marked if more than 80-word answer is given.]

calculated TOF is below the 3 mol per hour TOF reported for the same reaction (~12 mol%) (J. Chem. Educ. 2011, 88, 687) One probable cause could be partial catalyst oxidation due to air and moisture. This is supported by observed greenish colour in reaction mixture as well as $OPPh_3$ reported by ESI-MS.

END OF LAB PROFORMA

CM3191 (AY2026/27 Sem 1) – Inorganic Experiment 1

Homogeneous Catalysis: $RuCl_2(PPh_3)_3$ – Synthesis, Characterization and Catalytic Studies

APPENDICES to the Lab Proforma

KIM DONGHWAN (A0266094X) – Thursday group, fumehood B7
Sessions: I 20 Aug 2026, II 27 Aug 2026, III 3 Sep 2026

Appendix	Content	Pages
A	Datasheets, Sessions I–III (GTA-signed)	A-2 – A-4
B	ESI-MS spectra of $RuCl_2(PPh_3)_3$ (Session I) with peak assignments	A-5
C	ATR-IR spectrum of $RuCl_2(PPh_3)_3$ (Session I, Bruker, 500–4000 cm^{-1})	A-6
D	^1H NMR, Session II: full spectrum 0–10 ppm, all signals integrated, file name shown	A-7
E	^1H NMR, Session III: full spectrum 0–10 ppm, all signals integrated, file name shown	A-8
F	^1H NMR: expansions used for the quantitation (Sessions II and III)	A-9
G	Method note: referencing, integration and quantitation of the NMR spectra	A-10

Instruments and files. ESI-MS: Thermo ITMS, +p ESI, m/z 200–1400, 21 Aug 2026, file SN9_20260821131930. ATR-IR: Bruker (diamond ATR), 20 Aug 2026, file D:\CM3191\20Aug2026\hazel_Experiment1 Ru.3. ^1H NMR: Bruker 400.23 MHz, $CDCl_3$, zg30, NS 8, D1 1 s, SI 32768; Session II file CM3191-Thurs-400-Aug27-2026-09 (expno 1, procno 1); Session III file CM3191(Thurs)-400-Sep03-2026-09 (expno 1, procno 1). Tube number 09 in both sessions.

Reference values used. Mesitylene (internal standard, $CDCl_3$): CH_3 2.26 ppm (s, 9H), ArH 6.78 ppm (s, 3H) [SDBS; lecture notes p. 101]. Acetophenone: $COCH_3$ 2.58, ortho-H 7.96 ppm [SDBS; notes p. 100]. 1-Phenylethanol: CH 4.90 (q, J 6.5 Hz), CH_3 1.49 (d) [SDBS]. Diacetone alcohol: 1.27 (s, 6H), 2.20 (s, 3H), 2.63 (s, 2H) [SDBS]. Densities: 1-phenylethanol 1.012 g mL^{-1}, mesitylene 0.864 g mL^{-1} (supplier SDS). Molar masses: PPh_3 262.29, $RuCl_3$ 207.43 (anhydrous basis, notes p. 57), $RuCl_2(PPh_3)_3$ 958.85, 1-phenylethanol 122.17, mesitylene 120.19 g mol^{-1}.

Appendix A – Datasheet, Session I (20 Aug 2026)

DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 2

Session I: Synthesis and Characterization of $RuCl_2(PPh_3)_3$

Full name (as on student ID)	Matriculation no.	Date
KIM DONGHWAN	A0266094X.	2026/08/20, Thu.
Lab session ☐ Mon ☑ Thu	Fumehood no. B7.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
PPh_3 ✓	0.50 g	0.5076	
$RuCl_3 \cdot xH_2O$ ✓	0.08 g	0.080	
Dried absolute EtOH (to PPh_3)	20 mL	20	flask
Dried absolute EtOH (to $RuCl_3$)	20 mL	20	flask
Dried absolute EtOH (rinse)	5 mL	—	—
Cold dried EtOH (wash, 2 ×)	3 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Warming step 1: temp. / duration	~40 °C, 10–15 min		
Reflux start / end time	—		iPhone timer / after it started to drip
Reflux duration	90 min		~110°C.
Oil bath temp. / hotplate setting	< 130 °C	110–120	setting 200°C / 300 rpm.
Vacuum drying duration	20–30 min	26 mins.	

3. Product

	Value	Remarks
Mass of empty vial / paper	1.153 g.	
Mass of vial + dried product	1.395	
Mass of $RuCl_2(PPh_3)_3$	0.242 g.	
Colour of product	brown dark	reddish-brown color?
Mass submitted for ESI-MS		
IR method ☐ ATR ☐ KBr		

4. Observations

Record colour changes with the time at which they occurred: the two starting solutions, on mixing, during reflux, on cooling, of the filtrate and of the washings.

To be completed at the end of the session		
Student signature	GTA name	GTA signature 20/08/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

Appendix A – Datasheet, Session II (27 Aug 2026)

DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 3

Session II: Control Study — Oxidation *without* Catalyst

Full name (as on student ID)	Matriculation no.	Date
Lab session ☐ Mon ☐ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
Dried acetone	8 mL		
1-Phenylethanol	0.06 mL		
K_2CO_3	0.07 g	0.0704	
n-Hexane (quench / rinse 2×)	50 / 5 mL		
Mesitylene (internal standard)	0.07 mL		
$CDCl_3$	0.6 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Reflux start / end time	—		
Reflux duration	60 min		
Oil bath temp. / hotplate setting	< 100 °C		
Rotary evaporator bath temp. / pressure	—		
Vacuum drying of oil	15–20 min		

3. Conditions to reproduce in Session III

Write these down now. Next week's comparison depends on matching them.

Solvent batch / bottle	Ambient conditions

4. Observations

Colour on charging, during reflux and after cooling; appearance of the Celite pad and of the filtrate; appearance of the oily residue after concentration.

5. NMR sample

☐ Tube labelled with name, session and S/N no. ☐ Submitted S/N no. _____

To be completed at the end of the session		
Student signature	GTA name	GTA signature Song Tzu 27/08/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

Appendix A – Datasheet, Session III (3 Sep 2026)

DATASHEET — INORGANIC EXPERIMENT 1 — WEEK 4

Session III: Catalytic Oxidation *with* $RuCl_2(PPh_3)_3$

Full name (as on student ID)	Matriculation no.	Date
KIM. DongHwan	A0266094X.	
Lab session ☐ Mon ☑ Thu	Fumehood no.	S/N no. assigned by lab officer

1. Reagents charged

Reagent	Nominal	Actual	Remarks
Dried acetone (freed)	8 mL ✓		
1-Phenylethanol (fixed)	0.06 mL ✓		
K_2CO_3	0.07 g ✓	0.070	
$RuCl_2(PPh_3)_3$ (Week 2 product)	0.06 g ✓	0.060	
n-Hexane (quench / rinse 2×)	50 / 5 mL		
Mesitylene (internal standard) (fixed)	0.07 mL		
$CDCl_3$	0.6 mL		

2. Reaction conditions

	Nominal	Actual	Remarks
Reflux start / end time (1 phase final)	— ✓		
Reflux duration	60 min		
Oil bath temp. / hotplate setting	< 100 °C		
Vacuum drying of oil	15–20 min	15 min filtered w/ pump.	

3. Colour log — record the time with each entry

On adding the catalyst		Celite pad after filtration	
During reflux		First filtrate	
After cooling		Residue after rotary evaporation	

Second filtration required? ☐ No ☐ Yes — reason: _____

4. Deviations from the Session II conditions

add pump ✓ 5 min.

Solvent batch, ambient conditions, reflux duration, anything not matched

5. Other observations

· greenish color,

6. NMR sample

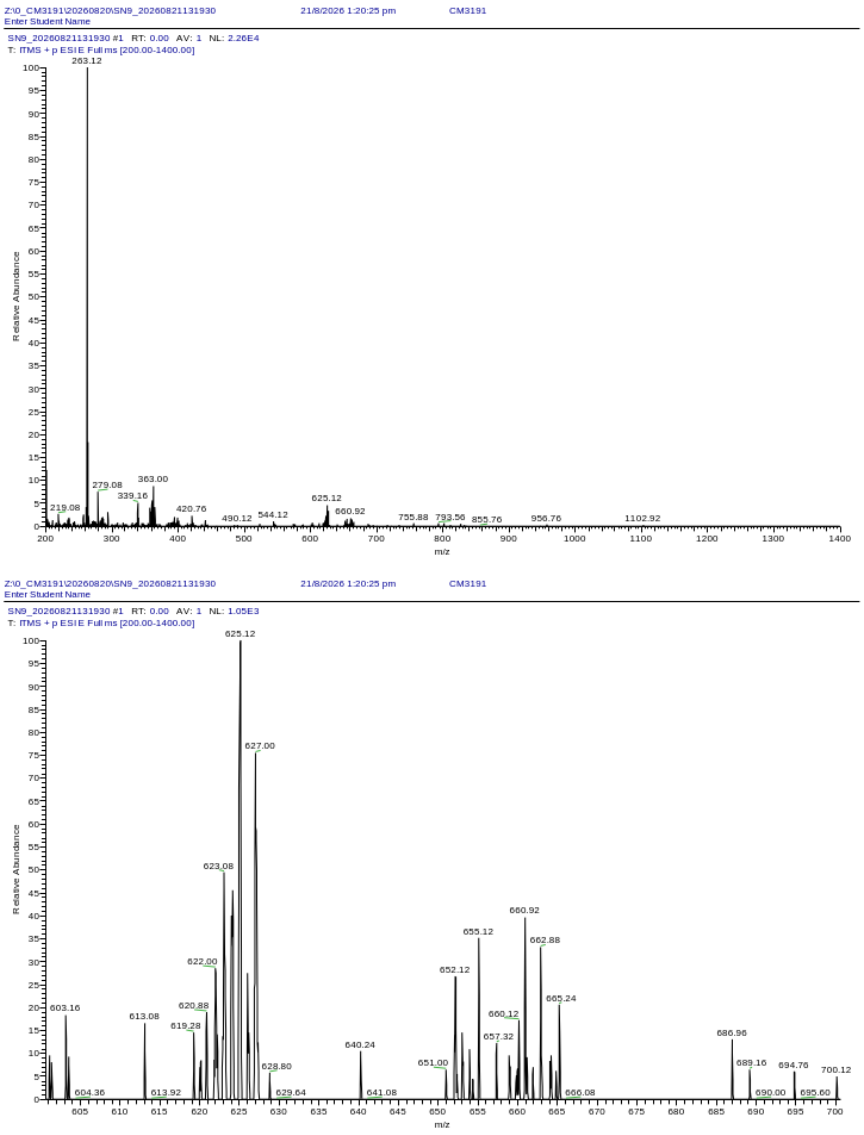
☐ Tube labelled with name, session and S/N no. ☐ Submitted S/N no. _____

To be completed at the end of the session		
Student signature	GTA name	GTA signature
		Sang Jun 3/9/2026

Raw data and observations must be signed by the GTA before leaving the laboratory. This datasheet is submitted as an appendix to the report.

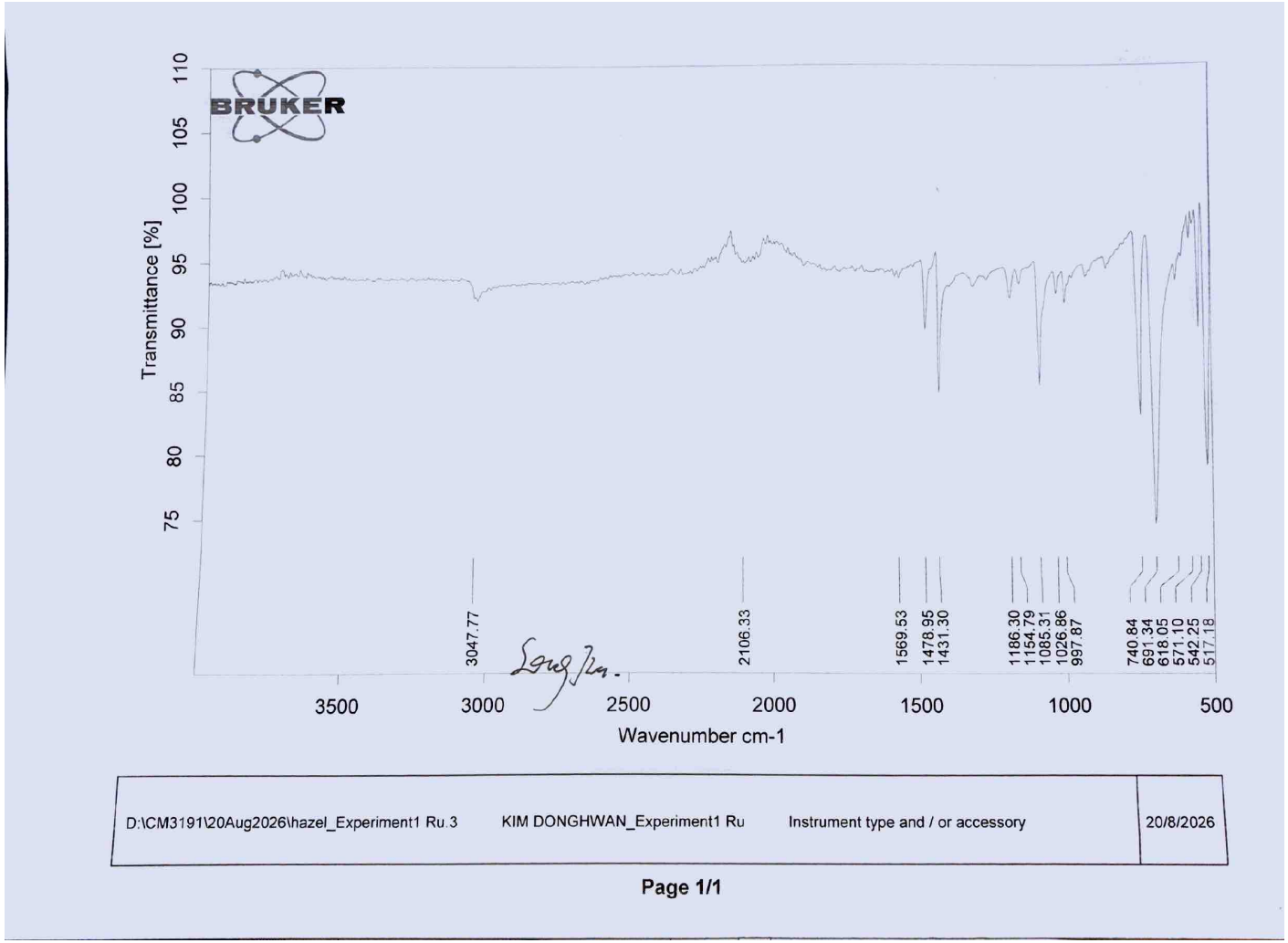
2026/09/03/Thu.

Appendix B – ESI-MS spectra of $RuCl_2(PPh_3)_3$ (Session I)



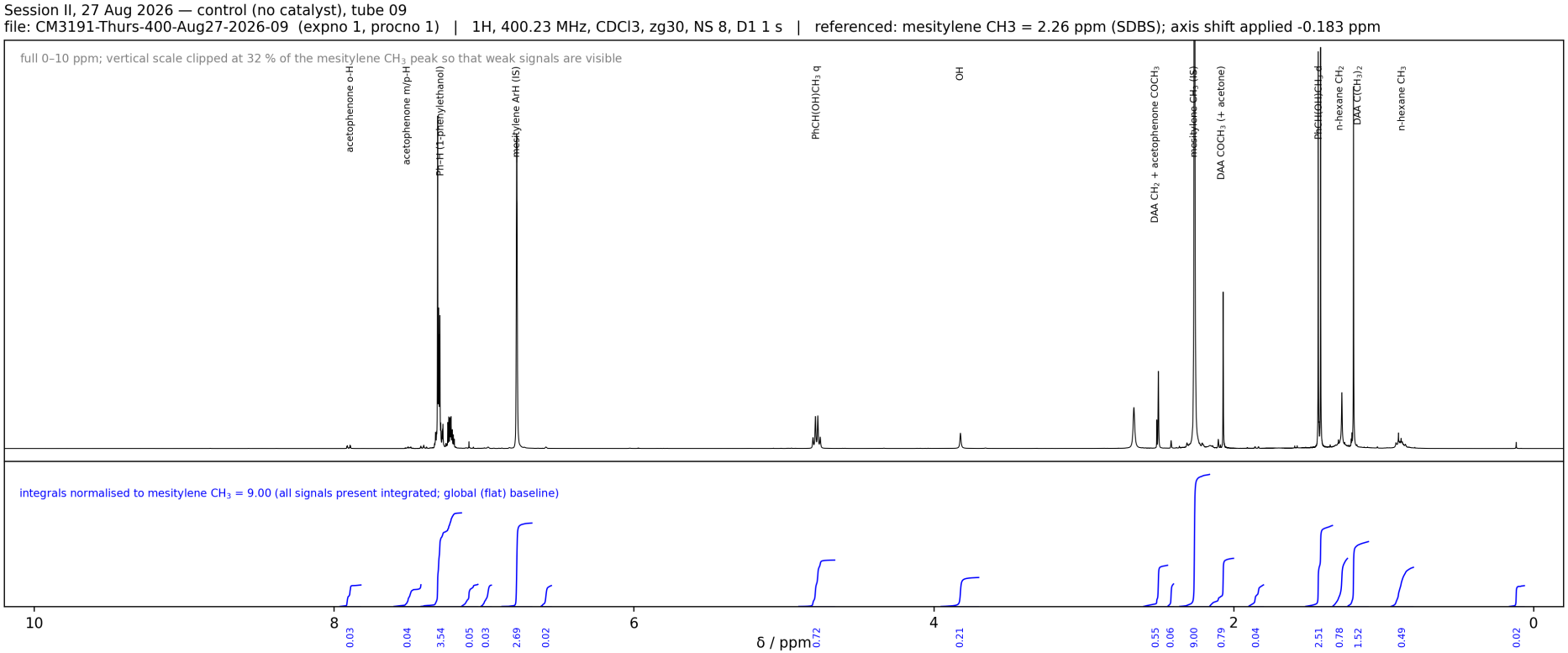
m/z obs.	Ion	m/z calc. (^{102}Ru, ^{35}Cl)	Note
263.12 (base)	$[PPh_3+H]^+$	263.10	free PPh_3 dissociated from the complex
279.08	$[OPPh_3+H]^+$	279.09	phosphine oxide by-product of the Ru(III)→Ru(II) reduction
339.16 (w)	$[PPh_4]^+$ (tentative)	339.13	phosphonium formed under ESI
619–629 (625.12)	$[Ru(PPh_3)_2-H]^+$ = $[RuCl(PPh_3)_2]^+ -$ HCl	625.08	Ru isotope envelope (∼8 u), no Cl
655–665 (660.92)	$[RuCl(PPh_3)_2]^+ = [M - Cl - PPh_3]^+$	661.06	Ru + Cl isotope pattern; confirms Ru–Cl–$(PPh_3)_2$ core
≈959, 923	$[M+H]^+$, $[M-Cl]^+$ not observed	–	PPh_3 lability and Cl^- loss under ESI

Appendix C – ATR-IR spectrum of $RuCl_2(PPh_3)_3$ (Session I, 20 Aug 2026)



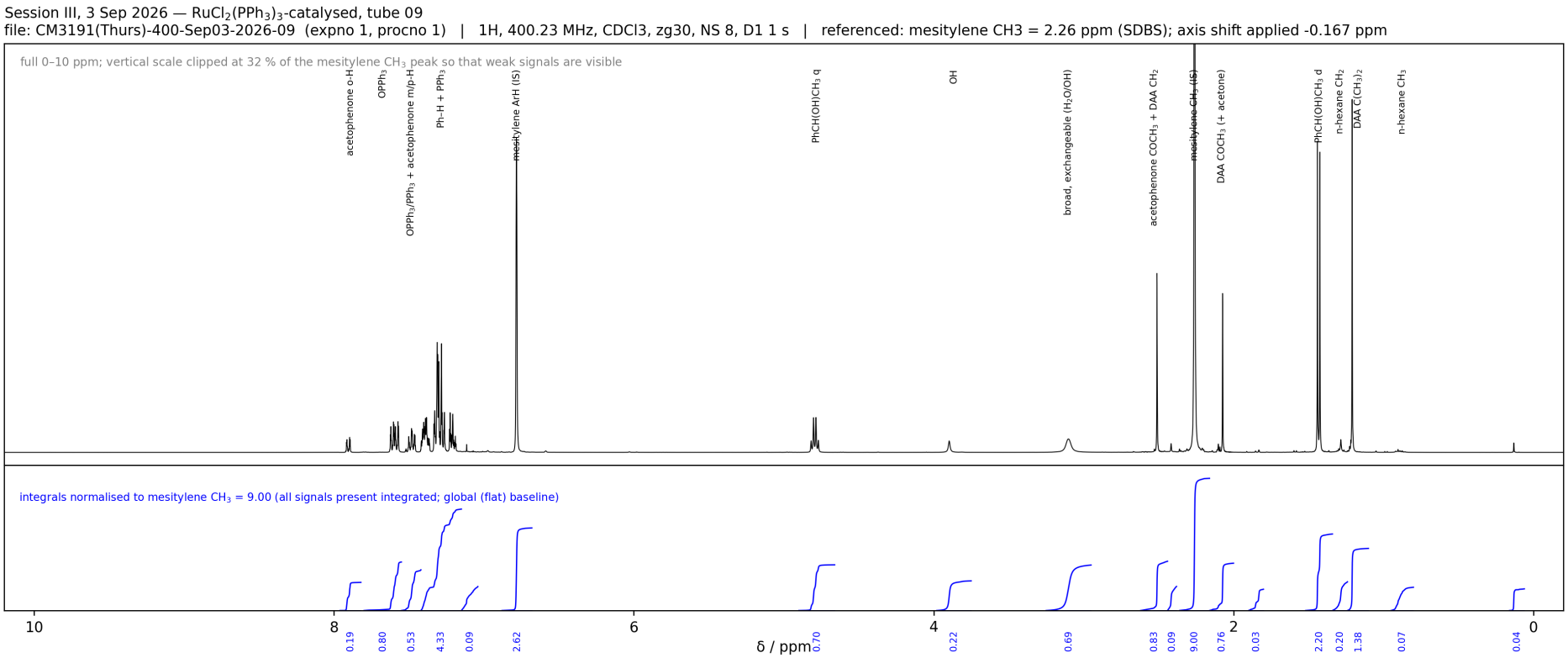
Bruker diamond-ATR, solid, 500–4000 cm^{-1}; file D:\CM3191\20Aug2026\hazel_Experiment1 Ru.3. Peaks (cm^{-1}): 3048, 2106, 1570, 1479, 1431, 1186, 1155, 1085, 1027, 998, 741, 691, 618, 571, 542, 517 – assigned in Q4 of the proforma.

Appendix D – ^{1}H NMR, Session II (control, no catalyst): full spectrum 0–10 ppm, all signals integrated



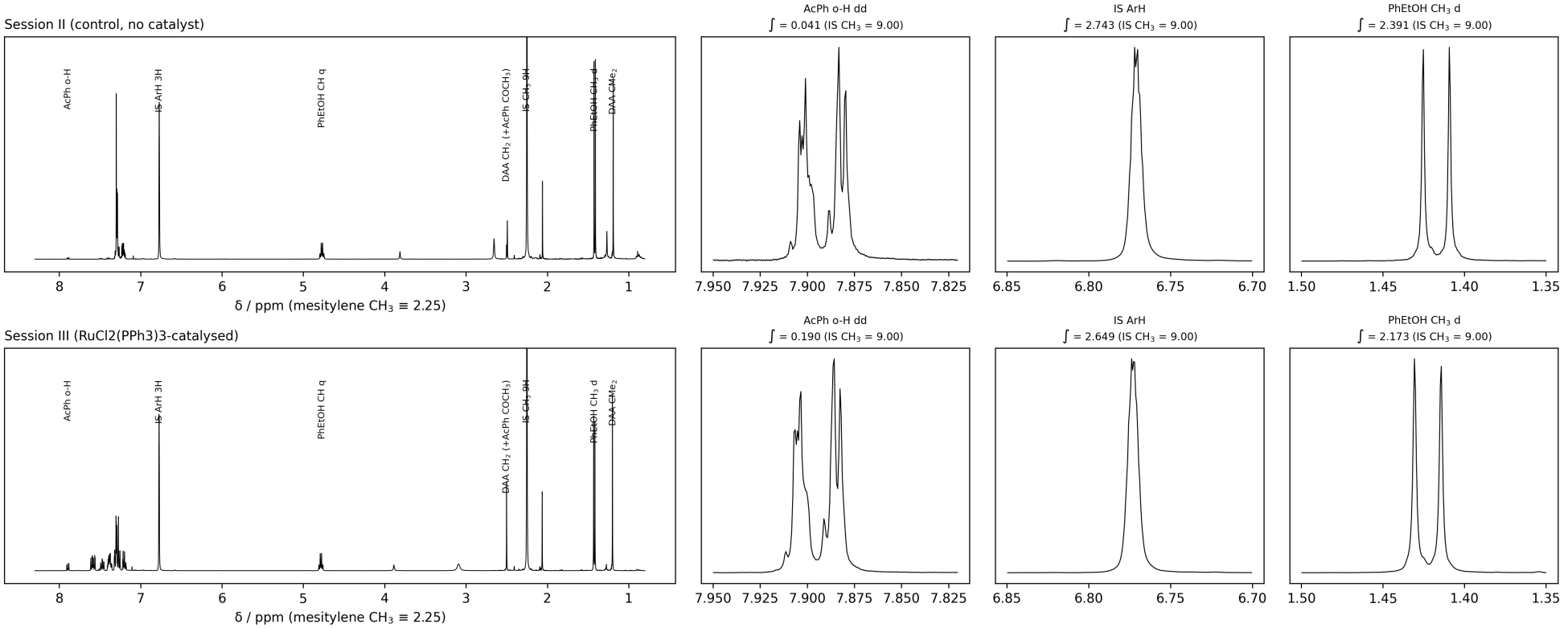
File CM3191-Thurs-400-Aug27-2026-09/1/pdata/1, tube 09, 27 Aug 2026. Mesitylene CH_3 set to 2.26 ppm (axis shift −0.183 ppm). Integrals normalised to mesitylene CH_3 = 9.00; integration windows (ppm): 7.96–7.82, 7.60–7.42, 7.42–7.15, 7.15–7.04, 7.02–6.95, 6.88–6.68, 6.62–6.55, 4.90–4.66, 3.95–3.70, 2.60–2.44, 2.44–2.40, 2.36–2.16, 2.16–2.00, 1.90–1.80, 1.52–1.34, 1.34–1.24, 1.24–1.10, 0.95–0.80, 0.16–0.06.

Appendix E – ^{1}H NMR, Session III ($RuCl_2(PPh_3)_3$-catalysed): full spectrum 0–10 ppm, all signals integrated



File CM3191(Thurs)-400-Sep03-2026-09/1/pdata/1, tube 09, 3 Sep 2026. Mesitylene CH_3 set to 2.26 ppm (axis shift −0.167 ppm). Same processing and windows as Appendix D, plus 7.80–7.55, 7.55–7.42 and 3.25–2.95 ppm. The additional aromatic multiplets (7.4–7.8 ppm) are PPh_3/$OPPh_3$ released from the catalyst.

Appendix F – ^{1}H NMR expansions used for the quantitation (Sessions II and III)



Left: full spectra after referencing (vertical scale clipped). Right: acetophenone ortho-H doublet (7.91/7.89 ppm), mesitylene ArH singlet (6.78 ppm) and 1-phenylethanol CH_3 doublet (1.44/1.42 ppm, J = 6.4 Hz) with integrals normalised to mesitylene CH_3 = 9.00.

Appendix G – Method note: referencing, integration and quantitation

1. **Reference.** No TMS was added and the residual $CHCl_3$ line (7.26 ppm) lies under the phenyl multiplet, so the mesitylene internal standard (manual, Sessions II/III step 12) was used as the chemical-shift reference as well: its CH_3 singlet was set to 2.26 ppm (SDBS value in $CDCl_3$, lecture notes p. 101). The spectrometer's automatic axis had placed it at 2.4425 (II) and 2.4266 ppm (III); the whole axis was shifted by -0.183 and -0.167 ppm respectively. After the shift the mesitylene ArH singlet falls at 6.78 ppm in both spectra, as expected. A constant shift does not change any peak area.

2. **Why the other signals are 0.05–0.13 ppm upfield of their SDBS values** (e.g. 1-phenylethanol CH 4.79 vs 4.90; acetophenone $COCH_3$ 2.51 vs 2.58): the samples are concentrated ($\sim$20 % v/v organics in 0.6 mL $CDCl_3$) and mesitylene is an aromatic co-solvent (aromatic-solvent-induced shift). Assignments rest on multiplicity and coupling (J = 6.4 Hz for the CH/CH_3 pair), on the diacetone-alcohol integral ratio 6 : 3 : 2, and on the presence/absence pattern between the two sessions.

3. **Integration.** Bruker processed data (1r) were used without re-processing; a flat baseline (median of the signal-free regions 3.4–3.7, 5.6–6.4, 8.4–9.5 ppm) was subtracted; every signal between 0 and 10 ppm was integrated over a window wide enough to include its tails (windows listed under Appendices D and E); all integrals were normalised to mesitylene CH_3 = 9.00.

4. **Quantitation** (lecture notes p. 103):

$$n(\mathrm{X}) = \frac{I(\mathrm{X})/N_{\mathrm{H}}(\mathrm{X})}{I(\mathrm{IS})/N_{\mathrm{H}}(\mathrm{IS})} \times n(\mathrm{IS}), \qquad n(\mathrm{IS}) = \frac{0.07\ \mathrm{mL} \times 0.864\ \mathrm{g\,mL^{-1}}}{120.19\ \mathrm{g\,mol^{-1}}} = 5.03 \times 10^{-4}\ \mathrm{mol},$$

$$n(\text{1-phenylethanol charged}) = \frac{0.06\ \mathrm{mL} \times 1.012\ \mathrm{g\,mL^{-1}}}{122.17\ \mathrm{g\,mol^{-1}}} = 4.97 \times 10^{-4}\ \mathrm{mol}.$$

   Acetophenone was quantified from its ortho-H doublet (2H, 7.91/7.89 ppm) against the mesitylene ArH singlet (3H, 6.78 ppm), because the acetophenone $COCH_3$ singlet (2.51 ppm) overlaps the diacetone-alcohol CH_2 singlet. Aromatic was paired with aromatic since, with D1 = 1 s and a 30° pulse, aromatic CH and CH_3 protons are partially saturated to different extents (mesitylene ArH/CH_3 per-proton ratio 0.90 and 0.87 instead of 1.00); pairing like with like cancels most of this bias. Cross-check by the notes' CH_3/CH_3 method after subtracting the diacetone-alcohol CH_2 contribution ($I(CMe_2)\times 2/6$): Session III $0.828 - 1.379/3 = 0.368 \Rightarrow$ 6.2×10^{-5} mol (12.4 %), in agreement with the ortho-H value 5.4×10^{-5} mol (10.8 %) within the uncertainty of the overlap correction. Remaining 1-phenylethanol: CH_3 doublet (3H) against mesitylene CH_3 (9H).

5. **Integrals used** (mesitylene CH_3 = 9.00): Session II – ArH 2.743, acetophenone o-H 0.041, 1-phenylethanol CH_3 2.391; Session III – ArH 2.649, acetophenone o-H 0.190, 1-phenylethanol CH_3 2.173. These give n(acetophenone) = 1.13×10^{-5} mol (2.3 %) and 5.41×10^{-5} mol (10.9 %), as used in Q6 and Q7.

6. **Uncertainty.** Moving each integration boundary by ± 0.01 ppm changes the acetophenone integrals by $<$5 % (III) and $<$10 % (II). Recovery of substrate carbon (ketone + remaining alcohol) is 83–84 % in both sessions, i.e. the loss is work-up loss, not reaction.